

LAB EXPERIMENT RECORD

Set 4: Product Inventory Management

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To visualize product inventory data using a bar chart, stacked bar chart, and scatter plot in R, and to build an interactive dashboard combining the bar chart and stacked bar chart using R Shiny.

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

Note: The experiment table gives only Product ID, Product Name, and Quantity Available. Questions 2 and 3 also need a product category and a price, so the R program uses the following values: Category – Electronics, Clothing, Electronics, Appliances, Clothing; Price (\$) – 20, 15, 18, 25, 12 (for Products A to E respectively).

PROGRAM 1: Bar Chart of Product Quantities

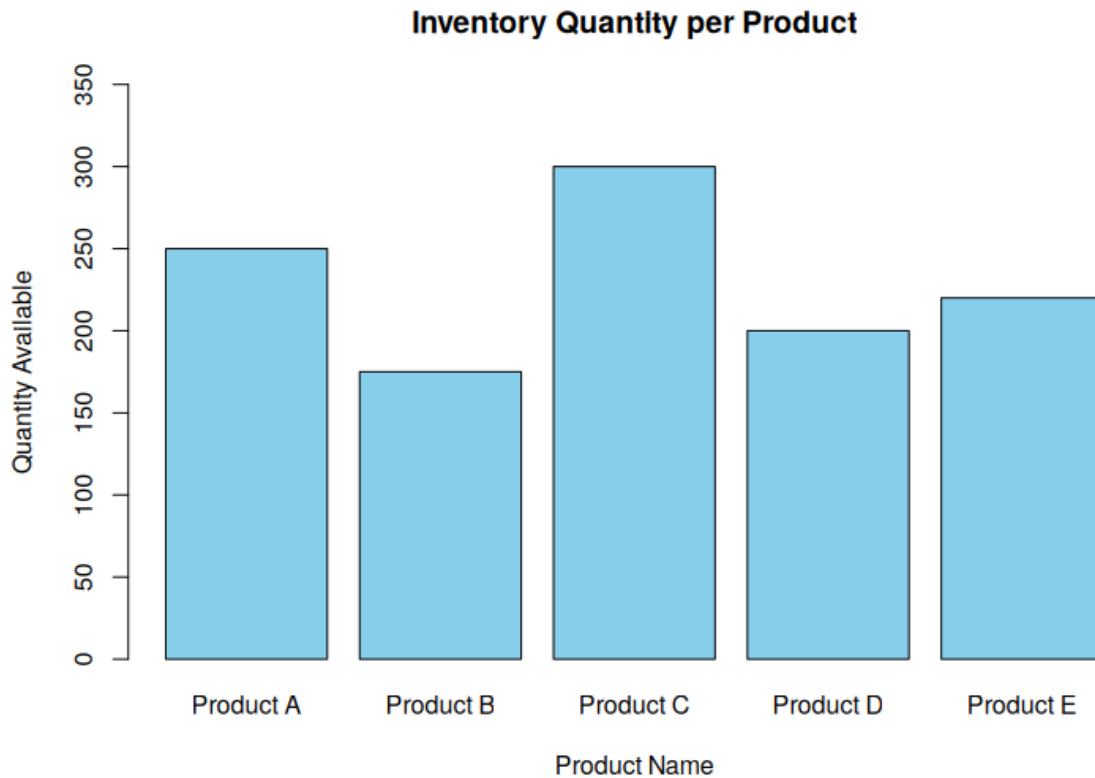
Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
# Create dataset for Set 4
inventory_data <- data.frame(
  Product_ID = c(1, 2, 3, 4, 5),
  Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Category = c("Electronics", "Clothing", "Electronics", "Appliances", "Clothing"),
  Quantity_Available = c(250, 175, 300, 200, 220),
  Price = c(20, 15, 18, 25, 12)
)

print(inventory_data)

# 1. Bar Chart for Product Quantities
barplot(inventory_data$Quantity_Available,
```

```
names.arg = inventory_data$Product_Name,  
col = "skyblue",  
xlab = "Product Name",  
ylab = "Quantity Available",  
main = "Inventory Quantity per Product",  
ylim = c(0, 350))
```



Output 1: Inventory Quantity per Product

Observation: Product C has the highest quantity in stock (300 units) and Product B has the lowest (175 units). Products A, D, and E lie in between with 250, 200, and 220 units.

PROGRAM 2: Stacked Bar Chart by Product Category

Generate a stacked bar chart to show the quantity of each product within different product categories.

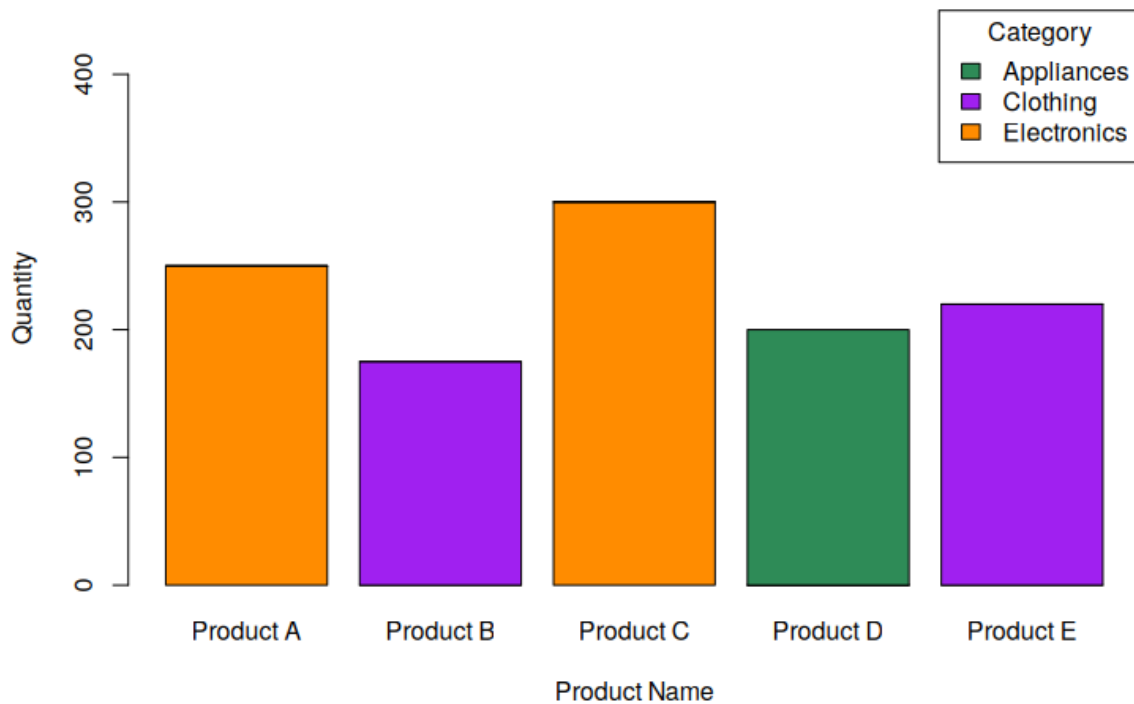
```
# Create dataset for Set 4
inventory_data <- data.frame(
  Product_ID = c(1, 2, 3, 4, 5),
  Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Category = c("Electronics", "Clothing", "Electronics", "Appliances", "Clothing"),
  Quantity_Available = c(250, 175, 300, 200, 220),
  Price = c(20, 15, 18, 25, 12)
)

print(inventory_data)

# Create contingency table of Product vs Category
category_table <- table(inventory_data$Category, inventory_data$Product_Name)

# Multiply by quantities to represent actual stock values
category_qty_matrix <- t(sapply(unique(inventory_data$Category), function(cat) {
  sapply(inventory_data$Product_Name, function(prod) {
    val <- inventory_data$Quantity_Available[inventory_data$Product_Name == prod &
inventory_data$Category == cat]
    if (length(val) == 0) return(0) else return(val)
  })
}))

# 2. Stacked Bar Chart
barplot(category_qty_matrix,
  col = c("darkorange", "purple", "seagreen"),
  xlab = "Product Name",
  ylab = "Quantity",
  main = "Product Quantities Stacked by Category",
  ylim = c(0, 450),
  legend.text = TRUE,
  args.legend = list(title = "Category", x = "topright"))
```

Product Quantities Stacked by Category**Output 2: Product Quantities Stacked by Category**

Observation: Electronics (Products A and C) holds the largest stock with 550 units, followed by Clothing (Products B and E) with 395 units and Appliances (Product D) with 200 units. Since each product belongs to only one category, every bar shows a single colour segment.

PROGRAM 3: Scatter Plot of Price vs Quantity Available

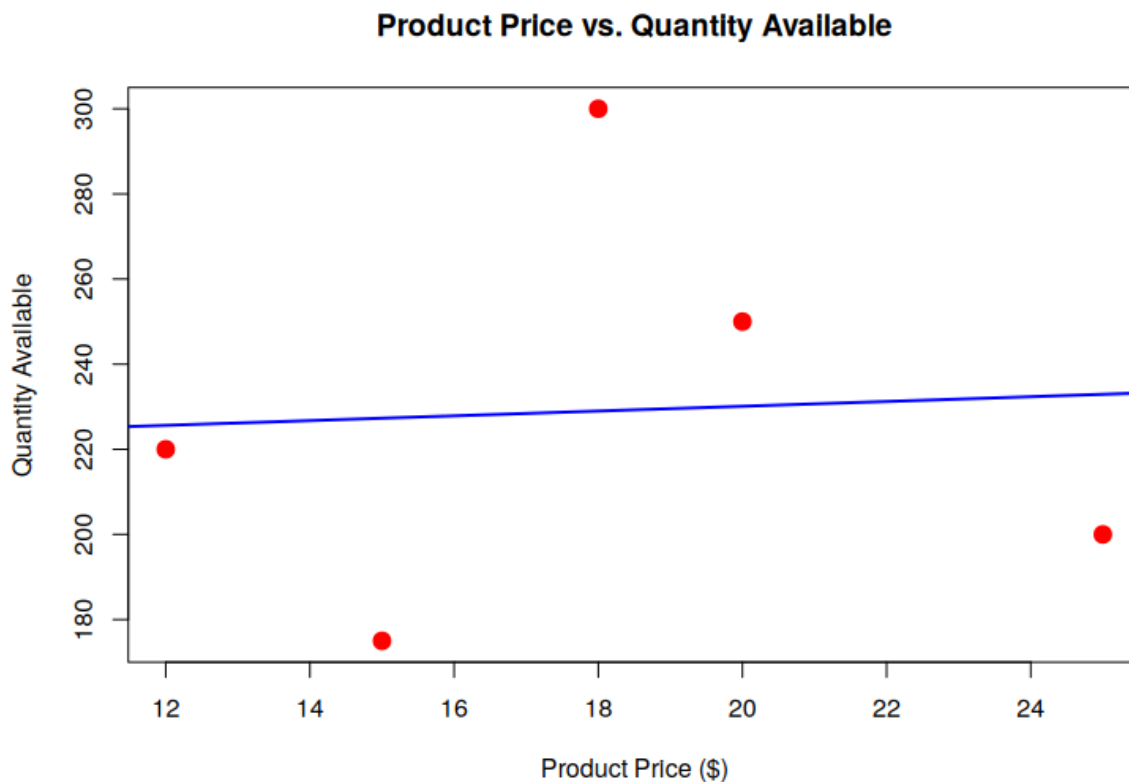
Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

```
# Create dataset for Set 4
inventory_data <- data.frame(
  Product_ID = c(1, 2, 3, 4, 5),
  Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Category = c("Electronics", "Clothing", "Electronics", "Appliances", "Clothing"),
  Quantity_Available = c(250, 175, 300, 200, 220),
  Price = c(20, 15, 18, 25, 12)
)

print(inventory_data)

# 3. Scatter Plot for Price vs Quantity Available
plot(inventory_data$Price, inventory_data$Quantity_Available,
     col = "red",
     pch = 19,
     cex = 1.5,
     xlab = "Product Price ($)",
     ylab = "Quantity Available",
     main = "Product Price vs. Quantity Available")

# Add a trend line
abline(lm(Quantity_Available ~ Price, data = inventory_data), col = "blue", lwd = 2)
```



Output 3: Product Price vs Quantity Available Scatter Plot

Findings: The points are scattered with no clear pattern and the trend line is almost flat (slope of about 0.56 units per dollar). The correlation coefficient is about 0.06, which means there is practically no relationship between price and quantity available. For example, Product C (\$18) has the highest stock while Product B (\$15) has the lowest. With only five products, this result is not conclusive.

PROGRAM 4: Interactive Dashboard for Inventory Data

Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 4
inventory_data <- data.frame(
  Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Category = c("Electronics", "Clothing", "Electronics", "Appliances", "Clothing"),
  Quantity_Available = c(250, 175, 300, 200, 220)
)
cat_cols <- c(Appliances = "seagreen", Clothing = "purple", Electronics = "darkorange")

ui <- fluidPage(
  titlePanel("Product Inventory Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("cat", "Category:",
        choices = names(cat_cols), selected = names(cat_cols))
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("barPlot", height = "340px")),
        column(6, plotOutput("stackPlot", height = "340px"))
      )
    )
  )
)

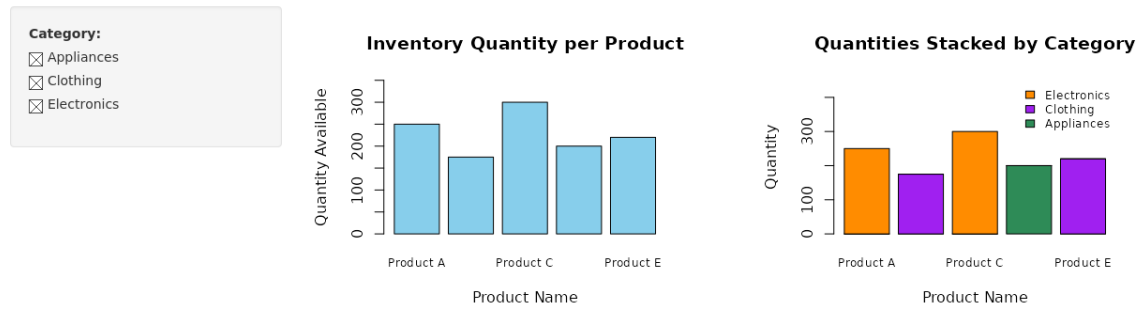
server <- function(input, output) {
  filtered <- reactive({ subset(inventory_data, Category %in% input$cat) })

  output$barPlot <- renderPlot({
    d <- filtered()
    barplot(d$Quantity_Available, names.arg = d$Product_Name, col = "skyblue",
      ylim = c(0, 350), xlab = "Product Name", ylab = "Quantity Available",
      main = "Inventory Quantity per Product", cex.names = 0.8)
  }, width = 420, height = 340, res = 96)

  output$stackPlot <- renderPlot({
    d <- filtered()
    m <- sapply(d$Product_Name, function(p)
      sapply(names(cat_cols), function(c)
        sum(d$Quantity_Available[d$Product_Name == p & d$Category == c])))
    barplot(m, col = cat_cols, ylim = c(0, 450), xlab = "Product Name", ylab = "Quantity",
      main = "Quantities Stacked by Category", cex.names = 0.8,
      legend.text = names(cat_cols),
      args.legend = list(x = "topright", bty = "n", cex = 0.8))
  }, width = 420, height = 340, res = 96)
}

shinyApp(ui = ui, server = server)
```

Product Inventory Dashboard



Output 4: Product Inventory Interactive Dashboard (R Shiny)

Observation: The dashboard shows the inventory bar chart and the category-wise stacked bar chart side by side. Unticking a category in the Category filter removes its products from both charts at once, making it easy to compare stock within selected categories.

Procedure

1. Create the inventory data frame with Product ID, Product Name, Category, Quantity Available, and Price.
2. Use `barplot()` to show the quantity of each product.
3. Build a Category $\times$ Product matrix with `sapply()` and use `barplot()` to draw the stacked bar chart with a legend.
4. Use `plot()` for the price vs quantity scatter plot and `abline()` with `lm()` to add a trend line.
5. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts.
6. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.

Result

The product inventory data was successfully visualized using a bar chart, stacked bar chart, and scatter plot in R, and an interactive R Shiny dashboard was built combining the bar chart and stacked bar chart.

Key R Functions Used

Function	Purpose
<code>data.frame()</code>	Creates the inventory dataset.
<code>barplot()</code>	Creates the bar chart and stacked bar chart.
<code>sapply()</code>	Builds the category-wise quantity matrix.
<code>plot()</code>	Creates the scatter plot.
<code>abline()</code> / <code>lm()</code>	Adds a linear trend line to the scatter plot.
<code>fluidPage()</code> / <code>checkboxGroupInput()</code>	Creates the dashboard layout and filters.
<code>reactive()</code> / <code>renderPlot()</code>	Filters the data and redraws the charts.
<code>shinyApp()</code>	Runs the interactive dashboard.